

Nikolas Alansson

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EDUCATION

Bachelor of Science in Chemistry and Biochemistry | July 2022 | University of Bremen, Germany

Master of Science in Chemistry | August 2024 | Georgia Institute of Technology, Atlanta, GA, USA

EXPERIENCE

PhD Student | Institute for Structural Biology, Grenoble, France | October 2024 - current | Research Directors: Yvain Nicolet & Sandrine Ollagnier de Choudens

- Research project: phylogenetic and biochemical characterization of the biotin synthase family
 - Anaerobic purification of proteins, development of purification protocols for proteins
 - Design and optimization of anaerobic activity assays with HPLC analysis
- Anaerobic structural characterization of proteins via X-Ray crystallography
 - Screening and optimization of crystallization conditions of proteins under anaerobic conditions
- Bioinformatic characterization of protein families

Graduate Research Assistant | Georgia Institute of Technology, Atlanta, USA | August 2022 – August 2024 | Research Director: Lynn Kamerlin

- Research project: Loop dynamics and mechanistic investigation of archaeal tyrosine phosphatases
 - Computational modelling of protein dynamics using molecular dynamics simulations
- Manuscript in preparation: “Loop Dynamics and Catalysis in Archaeal Protein Tyrosine Phosphatases”

Visiting Research Assistant | Utah State University, USA | May 2023 – June 2023

- Biochemical characterization and purification of artificial tyrosine phosphatases

Graduate Teaching Assistant | Georgia Institute of Technology | August 2022 - April 2024

- Train undergraduate students in entry-level chemistry experiments
- Explain complex scientific topics to undergraduate students during office hours
- Manage 45 first-year undergraduate students per semester

Undergraduate Research Assistant | Max-Planck-Institute for Marine Microbiology, Bremen, Germany | October 2020 - October 2021 | Research Director: Tristan Wagner

- Independent projects: structural and biochemical characterization of methylotrophic enzymes
- Refined x-ray crystallographic structures, both *ab initio* and through molecular replacement
- Anaerobic native purification of proteins, development and performance of activity assays
- First author manuscript in preparation - “Mechanistic insights into a methylotrophic methyltransferase”

Undergraduate Research Assistant | Center for Applied Space Technology and Microgravity, Bremen, Germany | October 2019 - October 2020 | Research Director: Cyprien Verseux

- Development of Gas Chromatograph analytical protocols for oxygen quantification of microbial space-analog cultures

Public Communications assistant | Center for Applied Space Technology and Microgravity, Bremen, Germany | February 2019 - October 2020

- Lead guided tours through research facility for general public groups, effectively and engagingly communicate concepts of microgravity and drop tower to public groups in German and English
- Safety officer and tour guide for public marriages on top of drop tower research facility

SKILLS & RELEVANT COURSEWORK

- Course: Graduate Teaching Assistant Preparation
- Proficient with crystallographic software packages – PHENIX, CCP4, BUSTER
- Experience with bioinformatic tools - MAFFT, MUSCLE, IQTREE, FASTTREE
- Proficient with molecular dynamics: GROMACS
- Proficient in protein expression, native and heterologous protein purification and crystallization
- Proficient in anaerobic protein expression, purification, crystallization
- Experience with HPLC systems
- Data Analysis: Python and R
- Science Communication and Education

LANGUAGES

- English – Native Speaker
- German – Native Speaker
- Icelandic – Elementary proficiency
- French - Elementary proficiency (A2)

GRANTS & AWARDS

2026 IMBG - IMBG Travel Grant

Travel grant for 2026 XII International Meeting for FeS Clusters in Gdansk, Poland

2026 EDCSV Doctoral School Day - Best Oral Presentation

- Poster Title: Journey to Understanding the Biotin Synthase Mechanism

2025 GT Sigma Xi - Best M.S. Thesis Award

- Thesis Title: Characterization Of Conformational Dynamics Of Archaeal Protein Tyrosine Phosphatases

2025 FrenchBIC - Poster Presentation Award

- Poster Title: Journey to Understanding the Biotin Synthase Mechanism

PUBLICATIONS

1. Shen R, Brownless A-L, **Alansson N**, Corbella M, Kamerlin SCL, Hengge A; SHP-1 Variants Broaden the Understanding of pH-Dependent Activities in Protein Tyrosine Phosphatases; JACS Au; 2024; <https://doi.org/10.1021/jacsau.4c00078>
2. Chad Pozarycki, Kenneth M. Seaton, Emily C. Vincent, Carlie Novak Sanders, Nickie Nuñez, Mariah Castillo, Ellery Ingall, Benjamin Klempay, Alexandra Pontefract, Luke A. Fisher, Emily R. Paris, Steffen Buessecker, **Nikolas B. Alansson**, Christopher E. Carr, Peter T. Doran, Jeff S. Bowman, Britney E. Schmidt and Amanda M. Stockton; Biosignature Molecules Accumulate and Persist in Evaporitic Brines: Implications for Planetary Exploration; Astrobiology, 2024; <https://doi.org/10.1089/ast.2023.0122>
3. Ramalho T, **Alansson N.**, Verseux C. (2020). "Cyanobacterial biomass production on Mars and Moon regolith and its utilization as a feedstock for other microorganisms and higher plants," *Proceedings of the MELiSSA Conference 2020*, Washington, DC.
4. Yehorova, D.; **Alansson, N.***; Shen, R.; Denson, J. M.; Robinson, M.; Risso, V. A.; Ramirez Molina, N.; Loria, J. P.; Gaucher, E. A.; Sanchez-Ruiz, J. M.; Hengge, A. C.; Johnson, S. J.; Kamerlin, S. C. L. Conformational Dynamics and Catalytic Backups in a Hyper-Thermostable Engineered Archaeal Protein Tyrosine Phosphatase. *JACS Au* 2026, 6 (1), 59–81. <https://doi.org/10.1021/jacsau.5c00756>. * **Shared-first authorship**

ORAL COMMUNICATIONS

FrenchBIC 2026 "Journey to Understanding the Biotin Synthase Phylogeny" Paris, France, 16-18 March, 2026

EDCSV Day 2026 "Journey to Understanding the Biotin Synthase Phylogeny" Grenoble, France, 17 June, 2026

XII International Conference on Fe/S Proteins: Function, Biogenesis and Regulation "Journey to Understanding the Biotin Synthase Phylogeny" **Gdańsk, Poland**, 30 September, 2026

POSTER COMMUNICATIONS

FrenchBIC 2025 "Journey to Understanding the Biotin Synthase Mechanism" Amboise, France, 19-22 May, 2025

COST Action Meeting FeSImmChemNet 2025 "Journey to Understanding the Biotin Synthase Mechanism" Belgrade, Serbia, 23 - 24 April 2025